SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA0305	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024 batch	**Date:**	13/08/26

Code of Conduct

I ___Harshavardhan. RV(192472163)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____HARSHAVARDHAN. RV____

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences
Failure to handle subject–verb agreement
Inefficient parsing for long queries
Example user query:
"Show me the transactions with the card from last month"
a) Analyze the ambiguity present in the given sentence using CFG
b) Evaluate the limitations of CFG in handling real-world conversational queries
c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:
Backtracking issues
Poor handling of incomplete or ambiguous input
Difficulty in real-time response
Example command:
"Book a flight to Delhi with a window seat"
a) Analyze how the given sentence can lead to multiple parse structures
b) Analyze the limitations of top-down parsing in this real-time application
c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs
d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.
The system must handle:
Complex and lengthy medical sentences
Ambiguity in sentence structure
Accurate subject-verb agreement
Relationships between medical actions and entities (sub-categorization)
Real-time processing for hospital use
Example input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.
a) Design a complete NLP architecture integrating:
CFG for syntactic structure
Probabilistic CFG (PCFG) for ambiguity resolution
Feature Structures for agreement handling
Efficient parsing techniques
b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).
c) Create a method to:
Incorporate sub-categorization frames for medical verbs
Handle real-time data processing
Improve accuracy and scalability in a hospital environment

ANSWERS

1. AI-Powered Customer Support Chatbot for Banking Services

a) Analyze the ambiguity present in the given sentence using CFG

Sentence: "Show me the transactions with the card from last month."

The sentence contains **structural ambiguity** because the phrases **"with the card"** and **"from last month"** can be attached in different ways.

A simple CFG representation is:

S → VP

VP → V NP

NP → N

NP → NP PP

PP → P NP

Possible Structure	Interpretation
[transactions [with the card] [from last month]]	Show transactions made using the card during last month
[transactions [from last month] [with the card]]	Show transactions from last month, with the card as additional information

Thus, CFG can generate multiple parse trees, but it cannot decide which meaning is most appropriate based on the banking context.

b) Evaluate the limitations of CFG in handling real-world conversational queries

Limitation	Explanation
Ambiguity resolution	CFG can generate multiple parse trees but cannot select the intended meaning
No probability	All grammatical interpretations are treated equally
Limited context	It does not consider banking context or previous conversation
Agreement handling	Basic CFG does not easily represent features such as number and agreement
Long queries	Parsing can become inefficient for long and complex sentences
Conversational input	Incomplete, informal, or ungrammatical user queries are difficult to handle

Therefore, basic CFG alone is not sufficient for an industry-level banking chatbot.

c) Evaluate and propose an improved solution using PCFG, Feature Structures, or Earley Parsing

A combined approach using **PCFG, Feature Structures, and Earley Parsing** can improve the system.

Method	Role	Improvement
PCFG	Assigns probabilities to grammar rules	Selects the most probable interpretation
Feature Structures	Stores grammatical features such as number	Improves agreement checking
Earley Parsing	Efficiently parses general CFGs	Handles long, ambiguous, and incomplete queries

PCFG: It selects the most likely parse using probabilities learned from banking queries.

Feature Structures: Example:

NP[number = plural]

VP[number = plural]

Earley Parsing: It handles complex and ambiguous customer queries efficiently and avoids unnecessary repeated parsing.

d) Justify how the proposed method improves accuracy and efficiency in an industry setting

Aspect	Improvement
Accuracy	PCFG selects the most probable interpretation
Grammar	Feature Structures improve agreement analysis
Efficiency	Earley Parsing handles complex queries efficiently
Context	Banking-domain probabilities improve interpretation
Customer Experience	Faster and more relevant responses are provided

Conclusion: Combining PCFG, Feature Structures, and Earley Parsing provides better accuracy and efficiency than basic CFG alone.

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2. Voice Assistant System

Example Command: “Book a flight to Delhi with a window seat.”

a) Analyze how the given sentence can lead to multiple parse structures

The phrase **“with a window seat”** creates ambiguity because it can be attached in different ways.

Parse Structure	Interpretation
Book [a flight to Delhi with a window seat]	Book a flight to Delhi and request a window seat
Book [a flight] [to Delhi with a window seat]	The phrase may be incorrectly associated with the destination phrase

Thus, multiple grammatical structures are possible, and the system must determine the user's intended meaning.

b) Analyze the limitations of top-down parsing in this real-time application

Top-down parsing starts from the start symbol and tries to generate the input using grammar rules.

Limitation	Explanation
Backtracking	Incorrect rule choices require the parser to return and try other rules
Ambiguity	Multiple interpretations create many parsing paths
Incomplete input	Partial spoken commands are difficult to process
Speed	Backtracking can increase response time
Prediction errors	The parser may predict structures that do not match the actual command
Real-time use	Delays can affect voice assistant performance

Therefore, top-down parsing alone is less suitable for real-time voice applications.

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

Earley Parsing uses three operations:

Operation	Function
Predictor	Predicts possible grammar rules
Scanner	Matches incoming words with grammar rules
Completer	Completes recognized grammatical structures

For the command:

“Book a flight to Delhi with a window seat”

Earley Parsing can maintain multiple possible interpretations while processing the input.

It can also process partial input such as:

“Book a flight to Delhi...”

and continue parsing when additional words are received.

Advantage	Benefit
Ambiguity handling	Maintains multiple possible parse structures
Partial input	Can continue when more words arrive
Efficiency	Avoids unnecessary repeated parsing
Incremental processing	Suitable for spoken input
Real-time response	Improves response speed

d) Compare and analyze the performance of Top-Down vs Earley Parsing

Feature	Top-Down Parsing	Earley Parsing
Parsing method	Starts from the start symbol	Uses Predictor, Scanner, and Completer
Backtracking	High	Reduced
Ambiguity handling	Difficult	Handles multiple parses efficiently
Partial input	Poor	Good
Real-time response	Less suitable	More suitable
Long sentences	Can be inefficient	More efficient
Industry use	Simple applications	Voice assistants and NLP systems

Conclusion: Earley Parsing is more suitable for Alexa/Siri-type applications because it handles ambiguity, incomplete input, and complex commands more efficiently.

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3. Healthcare Medical Report Analysis System

Example Input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

a) Design a complete NLP architecture

The proposed NLP architecture is:

Medical Report Input

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Preprocessing

(Tokenization and Sentence Splitting)

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Feature Structure Analysis

(Subject-Verb Agreement)

↓

CFG Parsing

(Syntactic Structure)

↓

PCFG

(Ambiguity Resolution)

↓

Earley Parsing

(Efficient Parsing)

↓

Sub-Categorization Analysis

(Verb–Entity Relationships)

↓

Medical Information Extraction

↓

Structured Output

Component	**Role in the System**
Preprocessing	Divides the medical report into sentences and tokens
CFG	Identifies the basic syntactic structure
PCFG	Resolves ambiguity using probabilities
Feature Structures	Checks subject–verb agreement and grammatical features
Earley Parsing	Efficiently parses long, ambiguous, and incomplete sentences
Sub-categorization Frames	Identifies relationships between verbs and medical entities
Information Extraction	Extracts diagnosis, treatment/action, and location

Thus, each component performs a specific task to improve the accuracy of medical information extraction.

b) Develop a step-by-step workflow from input to structured output

Step 1: Input

The system receives:

“The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

Step 2: Tokenization

The sentence is divided into tokens:

The | doctor | who | reviewed | the | patient | last | week |

recommends | starting | medication | and | scheduling |

a | follow-up | visit | in | Chennai

Step 3: CFG Parsing

The parser identifies:

Sentence Part	Identified Information
Subject	The doctor
Relative Clause	Who reviewed the patient last week
Main Verb	Recommends
Action 1	Starting medication
Action 2	Scheduling a follow-up visit
Location	Chennai

Step 4: Feature Structure Checking

The system checks subject–verb agreement:

Subject	Number	Verb
Doctor	Singular	Recommends

Step 5: PCFG Ambiguity Resolution

The phrase **"in Chennai"** may have different possible attachments.

PCFG selects the most probable interpretation:

Scheduling a follow-up visit → Chennai

Therefore, Chennai is identified as the location associated with the follow-up visit.

Step 6: Earley Parsing

Earley Parsing efficiently processes the complex sentence, including the embedded relative clause:

who reviewed the patient last week

This reduces repeated parsing and improves processing efficiency.

Step 7: Sub-Categorization Analysis

The system identifies relationships between actions and entities.

Verb	Related Entity/Action
Recommends	Starting medication
Recommends	Scheduling follow-up visit
Starting	Medication
Scheduling	Follow-up visit
Follow-up visit	Chennai

Step 8: Complete Workflow Table

Step	Process	Output
1	Input	Medical sentence received
2	Tokenization	Sentence divided into tokens
3	CFG Parsing	Basic syntactic structure identified
4	Feature Checking	Subject–verb agreement verified
5	PCFG	Most probable interpretation selected
6	Earley Parsing	Complex structure parsed efficiently
7	Sub-categorization	Verb–entity relationships identified
8	Information Extraction	Actions and location extracted
9	Structured Output	Final structured result generated

Final Structured Output

```
{
  "Diagnosis": "Not explicitly mentioned",
  "Actions": [
    "Start medication",
    "Schedule a follow-up visit"
  ],
  "Location": "Chennai"
}
```

c) Create a method to incorporate sub-categorization frames, handle real-time data processing, and improve accuracy and scalability

1. Incorporating Sub-Categorization Frames

The system can maintain a medical verb dictionary containing the expected relationships.

Medical Verb	Sub-Categorization Frame	Example
Diagnose	Doctor → Diagnosis → Patient	Doctor diagnoses diabetes
Prescribe	Doctor → Medication → Patient	Doctor prescribes medication
Recommend	Doctor → Action	Doctor recommends treatment
Start	Action → Medication	Start medication
Schedule	Action → Appointment/Visit	Schedule a follow-up visit
Review	Doctor → Patient/Report	Doctor reviews the patient report

For the given sentence:

RECOMMEND(Doctor, Start(Medication))

RECOMMEND(Doctor, Schedule(Follow-up Visit, Chennai))

This helps the system identify correct relationships between medical actions and entities.

2. Handling Real-Time Data Processing

The real-time processing workflow can be:

Streaming Medical Input

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Incremental Earley Parsing

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PCFG Ranking

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Feature Structure Checking

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Sub-Categorization Analysis

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Entity and Relationship Extraction

↓

Structured Medical Output

Requirement	**Method Used**
Real-time input	Incremental processing
Complex sentences	Earley Parsing
Ambiguity	PCFG
Agreement	Feature Structures
Medical relationships	Sub-categorization Frames
Fast output	Structured information extraction

3. Improving Accuracy

Method	**Accuracy Improvement**
Medical-specific PCFG	Selects likely medical sentence interpretations
Feature Structures	Checks grammatical agreement
Medical dictionaries	Identifies diseases, medicines, and treatments
Sub-categorization	Identifies correct verb–entity relationships
Confidence scores	Helps evaluate extracted information

4. Improving Scalability

Scalability Method	Benefit
Parallel processing	Processes multiple reports simultaneously
Earley Parsing	Efficiently handles complex sentences
Grammar caching	Reduces repeated computation
Sentence-level processing	Makes large reports easier to process
Structured databases	Supports efficient storage and retrieval
PCFG ranking	Focuses on the most probable parse

Final Conclusion

The proposed healthcare NLP architecture is more effective than basic CFG alone. CFG provides syntactic structure, PCFG resolves ambiguity, Feature Structures handle agreement, Earley Parsing improves efficiency, and sub-categorization frames identify relationships between medical actions and entities.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director

Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________